

THE STRUCTURE OF MATHEMATICAL REALITY

Phase III — Chapter 8

Vector Spaces and the Meaning of “Canvas”

Note Template

Self-Study Curriculum

How to use this template

Complete these notes **after** the reading and the lecture. The goal is not to copy phrases from the text — it is to reconstruct the ideas in your own language. Leave every item blank on first pass; fill it in from memory, then check.

1. A vector space is not a type of _____. It is a type of _____.

2. Three examples of vector spaces that are not columns of numbers

- a.
- b.
- c.

3. A basis is a choice of _____, not a fact about _____.

4. Coordinates change when you change _____. What does not change

list the invariants

5. A linear map preserves

Its matrix representation depends on:

Its eigenvalues do not depend on:

6. “Linearity is a property of the representation, not the system.”

My example to illustrate this:

choose one from your own work

7. An inner product defines

Changing the inner product changes:

8. The “right” coordinates/basis/inner product for a problem are

the ones that:

complete this sentence
